



Capital One Challenge Track

Autonomous Financial Intelligence & Resilience

1. About Capital One

Capital One is one of the most recognized brands in global banking, serving more than 150 million customers with a mission to change banking for good.

For nearly a decade, we have invested and innovated at the level of the world's leading technology companies. Today, we operate as an information-based technology enterprise backed by an in-house tech team. By combining a 100% cloud-native architecture, machine learning, and human-centric design, we improve our customers lives.

2. The Problem Today

Managing financial capital today remains fragmented, passive, and reactive across the entire economic ecosystem. Individual consumers suffer from decision fatigue and static credit models, missing opportunities to automate savings or demonstrate creditworthiness. Small business owners lack accessible treasury tools, leaving them vulnerable to sudden, unmanaged cash-flow crunches. Meanwhile, digital financial transactions face evolving fraud patterns that static rules fail to catch in real time.

3. The Big Challenge

How can we leverage real-time financial data and intelligent automation to build resilient tools that empower individuals, protect businesses, and safeguard digital capital?

4. Possible Approaches (Areas of Interest)

Teams are invited to tackle this challenge by focusing on **one** of three key areas:

1. **Consumer Financial Autonomy & Credit Building (B2C Focus):** Develop intelligent agents that transform raw transaction streams into active financial well-being—automating micro-savings, identifying subscription leaks, or establishing cash-flow-based credit scoring for thin-file consumers.
2. **SMB Cash-Flow & Working Capital Intelligence (B2B Focus):** Build predictive treasury engines for small businesses that aggregate revenue and operational expense streams to forecast 30-day liquidity, manage overhead, and recommend timely working capital buffers.
3. **Real-Time Anomaly & Security Sentinel (Risk & Security Focus):** Design behavioral anomaly detection engines that analyze transaction ledgers in real time to flag

unexpected transfer velocity, suspicious merchant category hops, or abnormal account behaviors.

5. Mentorship for Hackers

- **Capital One On-Site Technical Mentors:** Capital One software engineers and product managers available on-site throughout the 36-hour hackathon to assist with API integration, software design, and domain strategy. Hackers are encouraged to approach mentors to brainstorm and refine their ideas, troubleshoot technical hurdles, and ensure their projects effectively target the criteria in the judging rubric.

6. What to Build

A functional software prototype (web application, mobile app, API engine, or interactive dashboard) strongly encouraged to leverage the Capital One Nessie API as a resource to help teams avoid backend setup overhead, that solves a problem within one of the three areas of interest. Submissions must include a public code repository and a working live demo.

7. Why It Matters

Financial health is about more than just recording transactions; it is about giving people and businesses the agency to thrive. By building proactive, intelligent tools, hackers can help prevent small business failures, grant underserved populations access to credit, and establish safer digital banking ecosystems.

8. Judging Criteria

Submissions will be evaluated using standard 4-dimension judging criteria:

Metric		Focus	
Originality <i>How creative, unique, and fresh is the approach to solving the chosen problem?</i>	30 %	Substantiated Competitive differentiation	10%
		Identification of Market gap	10%
		Non trivial solution	10%
Technical Depth <i>How effectively does the project execute algorithmic/data logic within 36 hours?</i>	25 %	Data foundation	6%
		Algorithmic logic / Intelligence	9%
		System Design	5%
		Quality & functional demo	5%
Impact & Feasibility <i>Could this realistic prototype be adapted or deployed to protect or empower real-world banking users?</i>	25 %	Substantiated Business Model	10%
		Market Size(TAM/SAM/SOM)	5%
		Regulatory & Operational feasibility	5%
		Adoption strategy (Go-To-Market)	5%
Design & Experience <i>Is the interface intuitive, accessible, and engaging for its target user persona?</i>	20 %	Specific User Persona	7%
		Structured User Journey Map	7%
		Pitch	6%